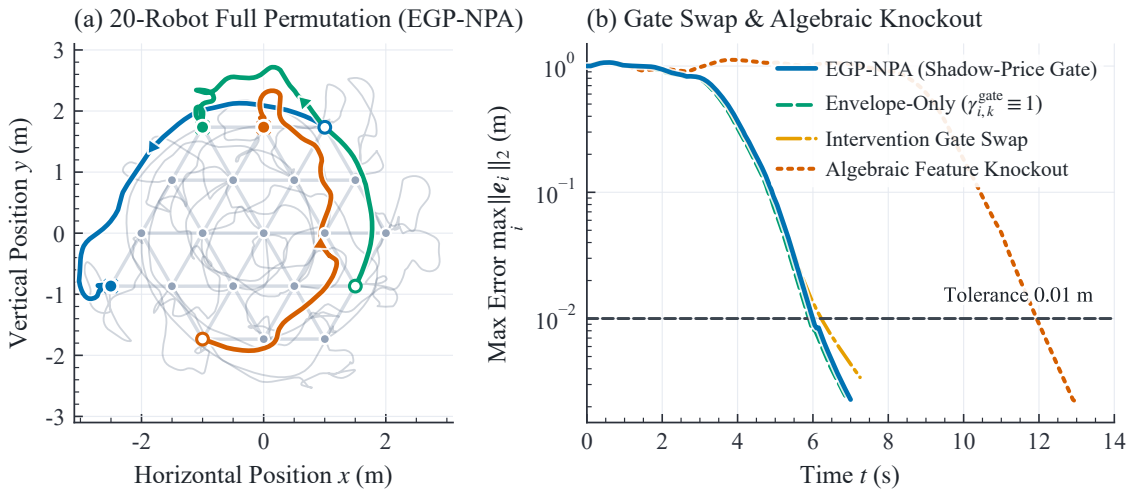

(a) 20-Robot Full Permutation (EGP-NPA)
Vertical Position y (m)
Horizontal Position x (m)
3
2
1
0
-1
-2
-3
-2
0
2
(b) Gate Swap & Algebraic Knockout
Max Error $\max_i \|e_i\|_2$ (m)
Time t (s)
10^0
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
EGP-NPA (Shadow-Price Gate)
Envelope-Only ($\gamma_{i,k}^{\text{gate}} \equiv 1$)
Intervention Gate Swap
Algebraic Feature Knockout
Tolerance 0.01 m